Cell-Type-Specific Biomarkers in Human Liver Fibrosis

Reproducible single-cell discovery pipeline, full report

2026-06-07

Overview

This report is generated by the liver-fibrosis pipeline. It summarizes QC, integration, annotation, donor-aware differential expression, pathway and cell-cell communication mechanism analysis, and the AI/ML biomarker prioritization that yields the ranked candidate table. Method rationale and citations are in `docs/methods.md`; written answers to the screening questions are in `docs/written_answers.md`.

QC summary

`### gse136103`

	sample	n_cells_before	n_cells_after	pct_kept	doublets_removed	mito_pct_ceiling
0	GSM4041150	1863	1478	79.3	4	8.00
1	GSM4041151	1039	655	63.0	0	8.00
2	GSM4041152	478	299	62.6	1	8.00
3	GSM4041153	6571	4762	72.5	1056	8.00
4	GSM4041154	4688	4128	88.1	1	8.00
5	GSM4041155	3295	2429	73.7	39	5.87
6	GSM4041156	3230	2550	78.9	4	7.06
7	GSM4041157	1156	820	70.9	5	8.00
8	GSM4041158	4937	4123	83.5	1	8.00
9	GSM4041159	3306	2826	85.5	1	6.37
10	GSM4041160	5122	4044	79.0	2	5.28
11	GSM4041161	2380	1764	74.1	9	6.35
12	GSM4041162	1686	1231	73.0	1	8.00
13	GSM4041163	2046	1231	60.2	3	8.00
14	GSM4041164	3412	2698	79.1	4	8.00
15	GSM4041165	3673	3031	82.5	2	6.19
16	GSM4041166	1985	1568	79.0	1	8.00
17	GSM4041167	3602	2955	82.0	0	7.64
18	GSM4041168	4819	3801	78.9	0	3.96
19	GSM4041169	2922	2061	70.5	5	5.17

`### gse244832`

	sample	n_cells_before	n_cells_after	pct_kept	doublets_removed	mito_pct_ceiling
0	JB288	7492	5650	75.4	87	0.60
1	JB289	3447	2152	62.4	12	0.11
2	JB290	9148	6693	73.2	10	0.57
3	JB303	8739	6764	77.4	28	0.32
4	JB304	6531	3702	56.7	35	0.00
5	JB305	7134	4408	61.8	19	0.10
6	JB317	6223	4350	69.9	5	0.06

	sample	n_cells_before	n_cells_after	pct_kept	doublets_removed	mito_pct_ceiling
7	JB318	6867	5158	75.1	2	0.43
8	JB319	7541	5447	72.2	4	0.18
9	JB320	8056	5793	71.9	3	0.08
10	JB321	7325	5432	74.2	0	0.17
11	JB322	8703	6863	78.9	5	0.56
12	JB336	4260	3360	78.9	23	0.28
13	JB337	7960	6320	79.4	5	0.20
14	JB338	7244	5412	74.7	2	0.17
15	JB339	6936	5308	76.5	10	0.35
16	JB340	7452	5835	78.3	6	0.34
17	JB341	7165	5327	74.3	1	0.34

Integration (scIB) and annotation

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	n_cells	n_batches	batch_silhouette	condition_silhouette	condition_knn_accuracy	fibrosis_signal_guard_pass
0	48454	20	-0.144	0.021	0.641	True

	leiden	n_cells	cell_type	compartment	frac_fibrotic_F3plus
0	0	7302	T_NK	lymphoid	0.499
1	1	6661	T_NK	lymphoid	0.287
2	2	6547	T_NK	lymphoid	0.301
3	3	5056	Monocyte_MoMac	macrophage_monocyte	0.502
4	4	4566	T_NK	lymphoid	0.347
5	5	3838	T_NK	lymphoid	0.354
6	6	3314	Endothelial_vasc	endothelial	0.722
7	7	1969	Endothelial_vasc	endothelial	0.642
8	8	1911	HSC_mesenchyme	stellate_myofibroblast	0.282
9	9	1758	Kupffer_resident	macrophage_monocyte	0.357
10	10	1462	Cholangiocyte	cholangiocyte	0.521
11	11	1350	LSEC	endothelial	0.250
12	12	1220	B_plasma	lymphoid	0.653
13	13	886	T_NK	lymphoid	0.221
14	14	237	Endothelial_vasc	endothelial	0.835
15	15	229	DC	myeloid_other	0.655
16	16	148	B_plasma	lymphoid	0.486

`### gse244832`

	n_cells	n_batches	batch_silhouette	condition_silhouette	condition_knn_accuracy	fibrosis_signal_guard_pass
0	93974	18	-0.176	-0.053	0.546	False

	leiden	n_cells	cell_type	compartment	frac_fibrotic_F3plus
0	0	15961	Hepatocyte	hepatocyte	0.501
1	1	15528	Hepatocyte	hepatocyte	0.439
2	2	12761	Hepatocyte	hepatocyte	0.435
3	3	12270	Hepatocyte	hepatocyte	0.633
4	4	8483	LSEC	endothelial	0.583
5	5	7028	Hepatocyte	hepatocyte	0.441
6	6	6761	LSEC	endothelial	0.590
7	7	6492	HSC_mesenchyme	stellate_myofibroblast	0.824
8	8	3364	Kupffer_resident	macrophage_monocyte	0.598
9	9	1607	Hepatocyte	hepatocyte	0.127
10	10	1309	HSC_mesenchyme	stellate_myofibroblast	0.906
11	11	861	Hepatocyte	hepatocyte	0.553

	leiden	n_cells	cell_type	compartment	frac_fibrotic_F3plus
12	12	754	Cholangiocyte	cholangiocyte	0.893
13	13	263	Endothelial_vasc	endothelial	0.992
14	14	219	Hepatocyte	hepatocyte	0.982
15	15	155	Hepatocyte	hepatocyte	0.129
16	16	98	Hepatocyte	hepatocyte	1.000
17	17	60	Hepatocyte	hepatocyte	1.000

Ranked biomarker / target candidates

The headline list is gated on differential-expression significance (compartment-level DESeq2 *or* a disease-associated-subcluster marker) and on cell-type specificity, so each candidate is a genuine, compartment-specific disease signal rather than an abundant lineage marker. Candidates flagged `in_disease_niche` are recovered from a disease-associated subpopulation that compartment-level pseudobulk averages out (e.g. scar-associated endothelium / macrophages).

	rank	gene	compartment	composite	log2FoldChange	padj	accessibility_class	druggability	in_disease_niche
0	1.0	PDGFRA	stellate_myofibroblast	0.671	3.608	0.000	surface	0.90	True
1	2.0	ACKR1	endothelial	0.635	4.059	0.000	surface	0.30	True
2	3.0	LUM	stellate_myofibroblast	0.624	2.771	0.000	secreted	0.10	True
3	4.0	DCN	stellate_myofibroblast	0.612	2.094	0.000	secreted	0.10	True
4	5.0	TREM2	macrophage_monocyte	0.606	0.694	0.028	surface	0.60	True
5	6.0	PLVAP	endothelial	0.581	0.239	0.376	surface	0.40	True
7	7.0	COL3A1	stellate_myofibroblast	0.553	3.555	0.000	secreted	0.20	True
9	8.0	MMP7	cholangiocyte	0.545	1.956	0.000	intracellular	0.10	False
10	9.0	SPP2	cholangiocyte	0.544	4.315	0.001	intracellular	0.10	False
11	10.0	MMP2	stellate_myofibroblast	0.542	3.328	0.000	secreted	0.55	True
12	11.0	COL1A1	stellate_myofibroblast	0.538	4.201	0.000	secreted	0.20	True
15	12.0	RGCC	endothelial	0.512	0.351	0.288	intracellular	0.10	True
16	13.0	COL1A2	stellate_myofibroblast	0.506	3.246	0.000	secreted	0.10	True
17	14.0	RGS4	cholangiocyte	0.495	4.156	0.001	intracellular	0.10	False
22	15.0	VWF	endothelial	0.477	2.782	0.000	surface	0.10	True
33	16.0	FABP4	endothelial	0.432	4.129	0.000	intracellular	0.10	True
34	17.0	RGS10	macrophage_monocyte	0.428	0.404	0.028	intracellular	0.10	True
49	18.0	IER3	macrophage_monocyte	0.406	0.939	0.000	intracellular	0.10	True
78	19.0	PLXDC2	macrophage_monocyte	0.390	0.315	0.106	intracellular	0.10	True
118	20.0	CTSH	macrophage_monocyte	0.362	0.771	0.001	intracellular	0.10	True

Known-positive recall (sanity check)

Does the pipeline recover biomarkers already established in the human-liver-fibrosis literature, and where a known positive is missed, *why* (non-significant / filtered / absent)? Panel: `resources/known_positive_markers.tsv`.

	metric	value
0	n_known_positives	37.00
1	recall_selected	10.00
2	recall_selected_frac	0.27
3	surfaced_as_candidate	25.00
4	detected_significant_in_primary	26.00
5	rescued_by_validation_only	1.00

	gene	compartment	found_status	rank	primary_padj	validation_status	rescued_by_validation
0	COL1A1	stellate_myofibroblast	selected	11.0	3.664619e-31	val_tested_nonsig	False
1	COL1A2	stellate_myofibroblast	selected	13.0	1.289774e-12	val_tested_nonsig	False
2	COL3A1	stellate_myofibroblast	selected	7.0	9.999196e-16	val_tested_nonsig	False
3	COL6A3	stellate_myofibroblast	DE_tested_nonsignificant	NaN	2.988873e-01	val_tested_nonsig	False
4	COL5A1	stellate_myofibroblast	scored_not_selected	NaN	3.243491e-08	val_tested_nonsig	False
5	ACTA2	stellate_myofibroblast	DE_tested_nonsignificant	NaN	4.781326e-01	val_tested_nonsig	False
6	TAGLN	stellate_myofibroblast	DE_tested_nonsignificant	NaN	5.233530e-01	val_tested_nonsig	False

	gene	compartment	found_status	rank	primary_padj	validation_status	rescued_by_validation
7	TIMP1	stellate_myofibroblast	scored_not_selected	NaN	1.927909e-11	val_tested_nonsig	False
8	PDGFRB	stellate_myofibroblast	DE_tested_nonsignificant	NaN	9.113876e-01	val_tested_nonsig	False
9	PDGFRA	stellate_myofibroblast	selected	1.0	7.830948e-10	val_tested_nonsig	False
10	PDGFD	stellate_myofibroblast	scored_not_selected	NaN	2.669875e-03	validated	False
11	LOXL1	stellate_myofibroblast	scored_not_selected	NaN	1.476092e-03	val_tested_nonsig	False
12	LOXL2	stellate_myofibroblast	scored_not_selected	NaN	7.423372e-03	val_tested_nonsig	False
13	TGFB1	stellate_myofibroblast	scored_not_selected	NaN	5.980219e-03	val_tested_nonsig	False
14	VCAN	stellate_myofibroblast	scored_not_selected	NaN	1.477132e-12	val_tested_nonsig	False
15	DCN	stellate_myofibroblast	selected	4.0	4.732443e-06	val_tested_nonsig	False
16	LUM	stellate_myofibroblast	selected	3.0	1.927909e-11	val_tested_nonsig	False
17	SPARC	stellate_myofibroblast	scored_not_selected	NaN	3.742002e-02	val_tested_nonsig	False
18	CCN2	stellate_myofibroblast	absent_primary	NaN	NaN	val_absent	False
19	FN1	stellate_myofibroblast	scored_not_selected	NaN	1.849887e-03	val_tested_nonsig	False
20	SMOC2	stellate_myofibroblast	DE_tested_nonsignificant	NaN	8.753296e-01	validated	True
21	THY1	stellate_myofibroblast	scored_not_selected	NaN	9.999196e-16	val_tested_nonsig	False
22	MMP2	stellate_myofibroblast	selected	10.0	2.200474e-09	val_tested_nonsig	False
23	TREM2	macrophage_monocyte	selected	5.0	2.765572e-02	val_absent	False
24	CD9	macrophage_monocyte	scored_not_selected	NaN	2.941762e-03	val_tested_nonsig	False
25	SPP1	macrophage_monocyte	scored_not_selected	NaN	1.362951e-01	val_tested_nonsig	False
26	GPNMB	macrophage_monocyte	DE_tested_nonsignificant	NaN	8.999708e-01	val_tested_nonsig	False
27	FABP5	macrophage_monocyte	DE_tested_nonsignificant	NaN	8.858496e-01	val_tested_nonsig	False
28	CD63	macrophage_monocyte	DE_tested_nonsignificant	NaN	9.924401e-01	val_tested_nonsig	False
29	LGMN	macrophage_monocyte	DE_significant_wrong_dir	NaN	2.087164e-02	val_tested_nonsig	False
30	PLVAP	endothelial	selected	6.0	3.761008e-01	val_tested_nonsig	False
31	ACKR1	endothelial	selected	2.0	1.200014e-07	val_absent	False
32	VWA1	endothelial	scored_not_selected	NaN	1.471575e-01	val_tested_nonsig	False
33	CD34	endothelial	DE_tested_nonsignificant	NaN	9.528367e-01	val_tested_nonsig	False
34	KRT19	cholangiocyte	scored_not_selected	NaN	9.376427e-01	val_tested_nonsig	False
35	KRT7	cholangiocyte	scored_not_selected	NaN	4.204185e-01	val_tested_nonsig	False
36	SOX9	cholangiocyte	DE_tested_nonsignificant	NaN	9.971689e-01	val_tested_nonsig	False

Niche (subpopulation) structure

Disease-associated subclusters within each compartment (donor-aware enrichment, valid within a sorted compartment); their one-vs-rest markers are the subset biomarkers folded into the ranked list above, recovering signals (PLVAP, TREM2, …) that compartment-level pseudobulk buries.

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	compartment	subcluster	n_cells	frac_test	frac_ref	disease_log2fc	disease_associated
9	stellate_myofibroblast	stellate_myofibroblast:8	50	0.1089	0.0035	4.613	True
44	lymphoid	lymphoid:10	95	0.0072	0.0003	2.710	True
23	endothelial	endothelial:10	236	0.0627	0.0124	2.249	True
32	endothelial	endothelial:9	291	0.0539	0.0116	2.119	True
27	endothelial	endothelial:4	647	0.1040	0.0269	1.911	True
21	endothelial	endothelial:0	1093	0.2446	0.0660	1.874	True
22	endothelial	endothelial:1	975	0.2054	0.0764	1.416	True
16	macrophage_monocyte	macrophage_monocyte:5	550	0.1167	0.0459	1.328	True
1	stellate_myofibroblast	stellate_myofibroblast:1	258	0.5364	0.2283	1.229	True
50	lymphoid	lymphoid:7	1130	0.0575	0.0258	1.124	True
37	cholangiocyte	cholangiocyte:4	191	0.1687	0.0799	1.069	True
31	endothelial	endothelial:8	305	0.0505	0.0255	0.958	True
8	stellate_myofibroblast	stellate_myofibroblast:7	95	0.1592	0.0892	0.829	True
15	macrophage_monocyte	macrophage_monocyte:4	753	0.1478	0.0835	0.816	True
34	cholangiocyte	cholangiocyte:1	246	0.2533	0.1565	0.691	True
46	lymphoid	lymphoid:3	4756	0.2080	0.1343	0.628	True

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	compartment	subcluster	n_cells	frac_test	frac_ref	disease_log2fc	disease_associated
8	stellate_myofibroblast	stellate_myofibroblast:8	220	0.4255	0.0000	8.737	True
38	hepatocyte	hepatocyte:8	219	0.0711	0.0007	5.372	True
20	endothelial	endothelial:10	266	0.0685	0.0037	3.890	True
29	endothelial	endothelial:9	437	0.0465	0.0048	3.023	True
36	hepatocyte	hepatocyte:6	1171	0.0331	0.0044	2.660	True
3	stellate_myofibroblast	stellate_myofibroblast:3	1085	0.1218	0.0507	1.249	True
4	stellate_myofibroblast	stellate_myofibroblast:4	943	0.1233	0.0535	1.189	True
16	macrophage_monocyte	macrophage_monocyte:6	135	0.0544	0.0246	1.111	True
28	endothelial	endothelial:8	470	0.0348	0.0209	0.707	True

Cross-dataset reproducibility

	gene	compartment	primary_lfc	primary_padj	valid_lfc	valid_padj	repro_score
0	COL1A1	stellate_myofibroblast	4.201	0.0	0.002	0.997	0.6
1	COL3A1	stellate_myofibroblast	3.555	0.0	0.018	0.929	0.6
2	RARRES1	stellate_myofibroblast	5.189	0.0	0.015	0.925	0.6
3	THY1	stellate_myofibroblast	3.737	0.0	0.015	0.851	0.6
4	SPON2	stellate_myofibroblast	3.678	0.0	0.018	0.949	0.6
5	MMP23B	stellate_myofibroblast	3.902	0.0	0.019	0.885	0.6
6	SERPINF1	stellate_myofibroblast	3.987	0.0	0.102	0.472	0.6
7	COL1A2	stellate_myofibroblast	3.246	0.0	0.038	0.813	0.6
8	VCAN	stellate_myofibroblast	5.709	0.0	0.182	0.320	0.6
9	LUM	stellate_myofibroblast	2.771	0.0	0.058	0.566	0.6
10	PCOLCE	stellate_myofibroblast	3.178	0.0	-0.013	0.947	0.0
11	TIMP1	stellate_myofibroblast	3.680	0.0	0.034	0.750	0.6
12	IGF1	stellate_myofibroblast	5.306	0.0	0.015	0.873	0.6
13	F2R	stellate_myofibroblast	3.704	0.0	0.072	0.550	0.6
14	PTGIS	stellate_myofibroblast	4.439	0.0	0.017	0.918	0.6
15	PDGFRA	stellate_myofibroblast	3.608	0.0	0.653	0.055	1.0
16	FBLN1	stellate_myofibroblast	3.314	0.0	0.009	0.966	0.6
17	MEG3	stellate_myofibroblast	2.530	0.0	-0.015	0.951	0.0
18	EFEMP1	stellate_myofibroblast	3.329	0.0	1.744	0.005	1.0
19	MMP2	stellate_myofibroblast	3.328	0.0	0.037	0.740	0.6

Mechanism: pathways and cell-cell communication

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	compartment	condition	source	mean_activity	level
0	cholangiocyte	cirrhotic	Androgen	1.0598	pathway_progeny
1	cholangiocyte	healthy	Androgen	0.8873	pathway_progeny
2	endothelial	cirrhotic	Androgen	0.4854	pathway_progeny
3	endothelial	healthy	Androgen	1.2176	pathway_progeny
4	lymphoid	cirrhotic	Androgen	0.3709	pathway_progeny
5	lymphoid	healthy	Androgen	0.3829	pathway_progeny
6	macrophage_monocyte	cirrhotic	Androgen	0.5556	pathway_progeny
7	macrophage_monocyte	healthy	Androgen	0.6782	pathway_progeny
8	myeloid_other	cirrhotic	Androgen	0.6113	pathway_progeny
9	myeloid_other	healthy	Androgen	0.6975	pathway_progeny
10	stellate_myofibroblast	cirrhotic	Androgen	0.2392	pathway_progeny
11	stellate_myofibroblast	healthy	Androgen	0.0296	pathway_progeny
12	cholangiocyte	cirrhotic	EGFR	2.2357	pathway_progeny
13	cholangiocyte	healthy	EGFR	1.9298	pathway_progeny
14	endothelial	cirrhotic	EGFR	2.4658	pathway_progeny

	source	target	ligand_complex	receptor_complex	fibrotic_condition	delta_specificity	enriched_in_fibrosis
0	myeloid_other	cholangiocyte	SCT	SCTR	cirrhotic	5.060615	True
1	cholangiocyte	lymphoid	DEFB1	CCR6	cirrhotic	4.847387	True
2	cholangiocyte	endothelial	CXCL6	ACKR1	cirrhotic	4.320696	True

	source	target	ligand_complex	receptor_complex	fibrotic_condition	delta_specificity	enriched_in_fibrosis
3	stellate_myofibroblast	endothelial	COL1A1	ITGA10_ITGB1	cirrhotic	4.167834	True
4	stellate_myofibroblast	endothelial	COL1A1	FLT4	cirrhotic	4.097689	True
5	stellate_myofibroblast	endothelial	COL1A1	ITGA5	cirrhotic	3.917435	True
6	stellate_myofibroblast	endothelial	COL3A1	ITGA10_ITGB1	cirrhotic	3.907577	True
7	stellate_myofibroblast	endothelial	COL1A1	CD93	cirrhotic	3.897793	True
8	stellate_myofibroblast	cholangiocyte	COL1A1	DDR1	cirrhotic	3.707833	True
9	stellate_myofibroblast	stellate_myofibroblast	COL1A1	ITGA1_ITGB1	cirrhotic	3.691121	True
10	stellate_myofibroblast	stellate_myofibroblast	COL1A1	DDR2	cirrhotic	3.682846	True
11	stellate_myofibroblast	cholangiocyte	COL1A1	ITGAV_ITGB8	cirrhotic	3.674624	True
12	stellate_myofibroblast	stellate_myofibroblast	COL1A2	ITGA1_ITGB1	cirrhotic	3.674624	True
13	stellate_myofibroblast	stellate_myofibroblast	THBS2	NOTCH3	cirrhotic	3.581461	True
14	endothelial	endothelial	CCL14	ACKR1	cirrhotic	3.581461	True

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	compartment	condition	source	mean_activity	level
0	cholangiocyte	MASH	Androgen	3.1403	pathway_progeny
1	cholangiocyte	MASL	Androgen	2.6690	pathway_progeny
2	cholangiocyte	normal	Androgen	1.8914	pathway_progeny
3	endothelial	MASH	Androgen	2.1593	pathway_progeny
4	endothelial	MASL	Androgen	2.2036	pathway_progeny
5	endothelial	normal	Androgen	2.4728	pathway_progeny
6	hepatocyte	MASH	Androgen	4.8672	pathway_progeny
7	hepatocyte	MASL	Androgen	4.7303	pathway_progeny
8	hepatocyte	normal	Androgen	4.5555	pathway_progeny
9	macrophage_monocyte	MASH	Androgen	2.2043	pathway_progeny
10	macrophage_monocyte	MASL	Androgen	1.8532	pathway_progeny
11	macrophage_monocyte	normal	Androgen	1.6571	pathway_progeny
12	stellate_myofibroblast	MASH	Androgen	2.5075	pathway_progeny
13	stellate_myofibroblast	MASL	Androgen	2.4753	pathway_progeny
14	stellate_myofibroblast	normal	Androgen	2.4165	pathway_progeny

	source	target	ligand_complex	receptor_complex	fibrotic_condition	delta_specificity	enriched_in_fibrosis
0	cholangiocyte	cholangiocyte	EFNA5	EPHA6	MASH	4.776138	True
1	stellate_myofibroblast	cholangiocyte	LAMA2	ITGA2_ITGB1	MASH	4.001292	True
2	endothelial	cholangiocyte	FGF23	FGFR2	MASH	3.972475	True
3	stellate_myofibroblast	cholangiocyte	FGF7	FGFR2	MASH	3.863243	True
4	stellate_myofibroblast	cholangiocyte	LAMA2	ITGA6_ITGB4	MASH	3.750497	True
5	cholangiocyte	stellate_myofibroblast	EFNA5	EPHA3	MASH	3.421758	True
6	endothelial	cholangiocyte	EFNB2	EPHA6	MASH	3.281674	True
7	cholangiocyte	cholangiocyte	EFNA5	EPHB6	MASH	3.266454	True
8	cholangiocyte	cholangiocyte	LAMC2	ITGAV_ITGB8	MASH	3.191933	True
9	stellate_myofibroblast	cholangiocyte	COL3A1	ITGA2_ITGB1	MASH	3.082785	True
10	stellate_myofibroblast	cholangiocyte	LAMC3	ITGA2_ITGB1	MASH	2.967011	True
11	endothelial	cholangiocyte	HSPG2	ITGA2	MASH	2.922234	True
12	stellate_myofibroblast	cholangiocyte	COL6A3	ITGA2_ITGB1	MASH	2.785301	True
13	stellate_myofibroblast	cholangiocyte	COL4A1	ITGA2_ITGB1	MASH	2.724684	True
14	cholangiocyte	stellate_myofibroblast	NCAM1	CACNA1C	MASH	2.711973	True